

CODE BOOK

for
ADP CARDS
of

CRUDE OIL ANALYSES

from
BUREAU OF MINES
OPEN FILES

Bartlesville Petroleum Research Center
Bureau of Mines
U.S. Department of the Interior
Bartlesville, Oklahoma

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EXPLANATION OF CODING ON ADP CARDS FOR CRUDE OIL ANALYTICAL DATA FROM U.S. BUREAU OF MINES ANALYSES

Card Number 1

Sample Location - Columns 1-5

These columns are used to designate the geographical and/or political subdivisions from which the sample was obtained. The codes used are given in the Appendix. The Appendix is arranged alphabetically according to continents or areas. All the states of the United States of America and provinces of the Dominion of Canada are coded numerically. Codes for 48 states of the USA are from the code book, "Numerical Code for States, Counties, and Cities of the United States," published in 1952 by International Business Machines Corporation. Other areas of the world are coded as noted in the Appendix. In most instances, the codes represent political subdivisions; however, where these were uncertain, unavailable, or too large, area or field designations have been used.

Sample Designation - Columns 6-11

Alphabetical codes in column 6 identify the laboratory or organization from which the crude oil analysis came. The cards include the Open File Analyses by the Bureau of Mines, some published analyses by the U. S. Geological Survey, and analyses of Canadian oils kindly furnished by Canadian governmental agencies. Table 1 shows the codes designating each organization.

TABLE 1. - Codes identifying laboratories and organizations - column 6

Code	Laboratory
A	Alberta, Canada - Oil and Gas Conservation Board of Alberta
B	Bartlesville, Oklahoma - U.S. Bureau of Mines
G	Casper, Wyoming - U.S. Geological Survey
L	Laramie, Wyoming - U.S. Bureau of Mines
O (alpha)	Ottawa, Canada - Mines Branch, Division of Fuels, Department of Mines and Technical Surveys

Numeric values in columns 7 and 8 identify, where possible, the year in which the sample was obtained or analyzed. The principal exceptions are some samples analyzed between 1918 and 1923 that do not have the year identification included in the sample number.

Numeric values in columns 9, 10, and 11 refer to the original sample number assigned by the laboratory. Information on the original sample numbers for specific analyses can be furnished upon request. Examples of the sample identification code are given in table 2.

TABLE 2. - Identification code

Code	Origin of data	Date of sampling
B 00 001	Bartlesville, Oklahoma	before 1923
B 53 062	Bartlesville, Oklahoma	1953
O 14 401	Ottawa, Canada	1914
O 41 902	Ottawa, Canada	1941
A 44 153	Alberta, Canada	1944

Depth of Productive Formation - Columns 12-16

Depth represents the depth from the surface to the top of the productive zone. If depths were unknown, no punches were made. If the productive zone is at the surface, the depth was punched as 00000.

Type of Formation - Columns 17 and 18

The type of productive formation, such as sand or limestone, is coded in these columns using the alphabetical codes shown in Table 3. Combination of types is denoted by combining the codes as illustrated at the bottom of the table. If neither the type nor name of productive formation was known, no punches occur in these columns. If the name of the formation was known, but the type was not, 00 was punched.

Geological Age - Columns 19, 20, and 21

In column 19, numeric codes are used to modify the time periods which are designated in columns 20 and 21. Thus "0" indicates undetermined; 1 indicates upper (or late); 4, middle; and 7, lower (or early).

TABLE 3. - Alphabetical codes for various types and combinations of types
of geological formations - columns 17 and 18

Code	Type of formation
AO	Limestone or lime
BO	Sandstone, sand, siltstone, quartzite
CO	Chert
DO	Shale
EO	Dolomite
FO	Conglomerate
GO	Igneous
HO	Chalk
JO	Basalt
KO	Serpentine
LO	Granite wash
MO	Metamorphics
NO	Salt
PO	Schist
QO	Breccia
RO	Anhydrite
SO	Clay
TO	Coal
AB	Sandy limestone
AE	Dolomitic limestone
BD	Sandstone and shale, sandy shale, shale and siltstone

Time periods or combinations of time periods are designated by alpha codes in columns 20 and 21, as shown in Table 4. If only one time period is involved, column 20 is used with an "O" in column 21. For combinations, the codes are combined and the second time period is entered in column 21. For example, Devonian-Silurian would be RS.

TABLE 4. - Numeric and alpha codes for various geological time periods

Code	Explanation	Code	Explanation
A	Quaternary	M	Triassic
B	Recent	NQ	Carboniferous ¹
C	Pleistocene	N	Permian
D	Tertiary	P	Pennsylvanian
E	Pliocene	Q	Mississippian
F	Miocene	R	Devonian
G	Oligocene	S	Silurian
H	Eocene	T	Ordovician
J	Paleocene	U	Cambrian
K	Cretaceous	V	Precambrian
L	Jurassic		

¹Although this term is now largely obsolete, it is found on some older analyses and thus a code is required.

Specific Gravity of Crude Oil - Columns 22, 23, and 24

Specific gravity is reported to three decimal places and the code is the numeric value for gravities less than 1.000. An overpunch in column 24 indicates that the specific gravity is 1.000 or greater, and the interpreter will type out an alpha code for column 24. The coding system is shown in Table 5.

TABLE 5. - Coding for specific gravity and for percentages of sulfur and nitrogen

Numeric Value				Numeric Value			
Specific Gravity	wt. % Sulfur	wt. % Nitrogen	Code	Specific Gravity	wt. % Sulfur	wt. % Nitrogen	Code
0.999	9.99	0.999	999	1.006	10.06	1.006	00F
1.000	10.00	1.000	00&	1.007	10.07	1.007	00G
1.001	10.01	1.001	00A	1.008	10.08	1.008	00H
1.002	10.02	1.002	00B	1.009	10.09	1.009	00I
1.003	10.03	1.003	00C	1.010	10.10	1.010	01&
1.004	10.04	1.004	00D	1.011	10.11	1.011	01A
1.005	10.05	1.005	00E	1.012	10.12	1.012	01B

Weight Percent Sulfur in Crude Oil - Columns 25, 26, and 27

Weight percent sulfur is reported to the second decimal place. Values greater than 9.99 percent are coded as shown in Table 5.

Coding was necessary for some of the older analyses in which sulfur values were reported as "less than 0.10 percent." These were coded as -10. This is in contrast with a sulfur value of 0.10 which is typed out as 010.

Weight Percent Nitrogen in Crude Oil - Columns 28, 29, 30

Weight percent nitrogen is reported to the third decimal place. In the few instances where the value exceeded .999 percent, it was coded in the manner shown in Table 5. Thus, when the interpreter types out a nitrogen value of 01H, it should be read 1.018.

Carbon Residue - Columns 31, 32, and 33

Weight percent Conradson carbon residue in crude oil required no special coding. It is reported to one decimal place, and the 3 columns accommodate values up to 99.9 percent. If the value was not determined, there are no punches in columns 31, 32, and 33.

Distillation Residue - Columns 34, 35, and 36

Volume percent distillation residue of the crude oil is reported to the first decimal place and the 3 columns can accommodate values up to 99.9 percent without special codes.

Pour Point - Columns 37 and 38

The pour point of the crude oil in degrees Fahrenheit is recorded in columns 37 and 38. Pour points of less than 5°F are coded by a negative punch in column 37, so that the interpreter types out -5 in columns 37 and 38. The few oils that had pour points over 99 are coded in a manner similar to that used for specific gravity. A few extremely heavy and viscous oils or bitumens were described on analysis reports as either semisolid or semifluid. Both descriptions were coded as SS. Table 6 illustrates the coding for pour points.

TABLE 6. - System for coding pour points

Pour point, °F	Code
Less than 5	-5
100	0&
105	0E
110	1&
125	2E
Semisolid or Semifluid	SS

Specific Gravities of Fractions - Columns 39 to 80

The specific gravities of fractions 2 through 15 are punched into the cards, using three columns for each fraction. (Space limitations dictated omission of one fraction, and the first fraction was chosen because it is absent from so many crude oils.) If specific gravities are greater than 0.999, the values are coded as shown in Table 5.

Card Number 2

Sample Identification - Columns 1-6

Columns 1-6 identify the sample number which is identical to the sample number on Card 1. Sample identification numbers relate data on Card 2 to those on other ADP cards and to the original analysis report. Refer to Table 1, page 1, for identification of sample.

Volume Percent of Crude Oil in Each Fraction (1 through 15) - Columns 7-51

The volume percent of crude in each fraction is punched using three columns per fraction. The percent is reported to one decimal place, and the three columns accommodate values up to 99.9 percent.

Viscosities of Fractions 11 through 15 - Columns 52-64

Saybolt Universal viscosity at 100°F is reported for fractions 11-15. Two columns each were allowed for fractions 11 and 12; three columns each for fractions 13-15. Coding was necessary for viscosity reported as "less than," "more than," and for values with too many digits for the space allowed. Fractions 11 and 12 are reported in two columns, so a viscosity value of "less than 30" would be coded -3. This coding was not necessary for fractions 13, 14, and 15 where three columns are used, and a value of "less than 30" would be reported as -30. "Over" or "greater than" values are coded with a plus in the last column, so "over 400" would be 40+. Some machines interpret the 40+ as 40&. (Viscosities greater than 400 SUS were reported as "over 400" on many analysis reports.) Although viscosities of highly viscous oil have often been reported simply as "over 400," some analyses reports show measurement of higher viscosities. These 4-digit numbers require codes to fit the space allowed. Codes for viscosities of fractions are shown in Table 7.

Cloud Points of Fractions - Columns 65-77

Cloud points normally are reported as specific numerical values in degrees Fahrenheit unless they were less than 5°F. They normally are shown on the analysis report as "less than" or "below 5" whenever less than 5°F. However, some were reported as less than 0 on some analysis reports. Others were reported as negative values, such as 10° below zero or 20° below zero F on a few analysis reports. Values of "less than 5" were coded as -5, values of "less than 0" were coded as -0; and all negative values were coded as -0.

Values with three digits are shown in some analysis reports. These are not coded for fractions 13, 14, and 15 because the fractions are allowed three columns or three digits each. In extremely rare cases, three digit values for either fraction

TABLE 7. - Codes for viscosities of fractions, SUS at 100°F, columns 52-64

Fractions 11 and 12		Fractions 13, 14, and 15	
Viscosity	Code	Viscosity	Code
Less than 30	-3	Less than 30	-30
99	99	Less than 32	-32
100	T0	Over 400	40+ (40 &)
101	T1	999	999
105	T5	1000	A00
110	A0	1010	A01
120	B0	1020	A02
130	C0	1060	A06
140	D0	1100	A10
150	E0	1160	A16
160	F0	1850	A85
170	G0	2000	B00
180	H0	2140	B14
190	J0	3000	C00
200	K0	4000	D00
210	L0	5000	E00
220	M0	6000	F00
230	N0	Over 6000	F + 0 (F&0)
240	P0		
250	Q0		
260	R0		
270	S0		
280	U0		
290	V0		
300	W0		
310	X0		
320	Y0		
330	Z0		

11 or fraction 12 would require coding because these fractions are allowed two columns each on Card 2. Codes for cloud points are shown in Table 8.

TABLE 8. - Codes for cloud points of fractions, degrees F, columns 65-77

Fractions 11 and 12		Fractions 13, 14, and 15	
Cloud point	Code	Cloud point	Code
All negative temperatures	-0	All negative temperatures	-0
Less than 0	-0	Less than 0	-0
Less than 5	-5	Less than 5	-5
100	T0	100	100
105	T5	105	105
110	A0	110	110
115	A5	115	115
120	B0	120	120
125	B5	125	125
(Higher values coded as in Table 7)			

Crude Oil Viscosity at 100°F - Columns 78-80

Viscosities are reported in three columns, 78, 79, and 80, on Card 2. Most values require two or three digits, but a considerable number of analysis reports show values with four digits, and a few with five digits. Special coding, similar to that used for viscosities of fractions, is used for values over 999. Coded values report the viscosities to the nearest 10 seconds, Saybolt Universal Seconds.

Some crude oil viscosities are reported as "less than 32." They are coded as -32. If reported as "less than 30," the values would be coded as -30. Codes for viscosities of crude oils are shown in Table 9.

TABLE 9. - Codes for viscosity of crude oil, SUS at 100°F, columns 78-80

Viscosity	Code	Viscosity	Code
Less than 30	-30	7000	G00
Less than 32	-32	8000	H00
999	999	9000	I00
1000	A00	10000	J00
1010	A01	11000	K00
1020	A02	12000	L00
2000	B00	13000	M00
3000	C00	Over 13000	M + 0 (M & 0)
4000	D00	Over 25000	Y + 0 (Y & 0)
5000	E00	Solid at 100°	000
6000	F00		
Over 6000	F+0 (F & 0)		

Card Number Zero

Card zero is basically an "identification" card. It contains: (1) analysis number; (2) the same geographical location code that is on card 1; (3) literal field name; (4) geographical code that is on card 1; (5) literal geological name of producing formation; (6) the American Association of Petroleum Geologists Stratigraphic Code (AAPG Code) for producing formation; and (7) a one-column code for modifying the literal geological code.

Sample Location - Columns 1-5

State and county, or foreign area code, as explained for columns 1-5 for card number 1.

Sample Number - Columns 6-11

Identical to columns 6-11 for card number 1.

Literal Field Name - Columns 12-44

The most commonly accepted name for the field, except that East, West, North, South, Northwest, etc., does not precede the field name being modified except for East Texas. All other direction-modified names are followed by a comma and the accepted direction abbreviation (without spacing or period), e.g., Edmond,W;

Edmond,NW; Pass,S; etc. Qualifiers such as named areas within a field are entered within parentheses following the field name when sufficient columns are available. Logical abbreviation of this information is sometimes used.

Geological Age of Producing Formation - Columns 45-47

Identical to columns 19-21 for card 1.

Geological Name of Producing Formation - Columns 48-70

The most commonly accepted literal spelling of the geological name. The smallest named subdivision is entered first, followed by the next smallest, e.g., Winger, Strawn. Upper, Middle, and Lower (or equivalent) are not entered.

Geological names for which numbers have been assigned in the AAPG Code are entered only when names of their subdivisions are not entered. Following is an alphabetical listing of the names:

Alexandrian	Guadalupe	Paleocene
Atoka	Gulf	Paleozoic
Cambrian	Jurassic	Pennsylvanian
Canadian	Kinderhook	Permian
Cayugan	Leonard	Pleistocene
Cenozoic	Meramec	Pliocene
Champlainian	Mesozoic	Precambrian
Chautauquan	Miocene	Quaternary
Chester	Mississippian	Recent
Cincinnatian	Missouri	Senecan
Coahuila	Morrow	Silurian
Comanche	Niagaran	Tertiary
Cretaceous	Ochoa	Triassic
Des Moines	Oligocene	Ulsterian
Devonian	Ordovician	Virgil
Eocene	Osage	Wolfcamp
Erian		

If the geological name in the preceding list is the only known formation name, it is entered and an appropriate entry is always made in column 71 as explained in the following section.

Modification of Geological Name - Column 71

This modification code serves two purposes: (1) To allow a programed printout of the geological name (columns 48-70), and the AAPG geological name, without

redundancy in the printout; and (2) to allow inclusion of Upper, Middle, and Lower in the printout as needed. Column 71 is coded as follows:

<u>Code</u>	<u>When Used</u>
No entry	If literal name is other than those for which numbers have been assigned in the AAPG Code.
1	Literal name not included in AAPG Code, and modified as upper.
2	Literal name included in AAPG Code and modified as upper.
3	Literal name included in AAPG Code and unmodified.
4	Literal name not included in AAPG Code and modified as middle.
5	Literal name included in AAPG Code and modified as middle.
6	Literal name included in AAPG Code and modified by depth, letter, number, etc. If the literal name is also modified by upper, middle, or lower, <u>code 6 is not used</u> as codes 2, 5, and 7 are given precedence.
7	Literal name not included in AAPG Code and modified as lower.
8	Literal name included in AAPG Code and modified as lower.
9	No geological information.

Assignment of codes 1, 4, and 7 for upper, middle, and lower, respectively, corresponds to assignment made by the AAPG Code. Codes 2, 5, and 8 (increment of one to 1, 4, and 7) were logical assignments to indicate upper, middle, and lower, respectively, when the literal name corresponds to a name for which a number has been assigned in the AAPG Code. This differentiation is desirable as explained in the examples in the following section regarding the AAPG Code.

AAPG Code

The AAPG Code is an eight-column code. The first three columns are numbers assigned to time-stratigraphic units in which the formation occurs (or actually is the unit). The next four columns contain an alphabetic mnemonic code for the name, and the last column is for modifying information.

Formations believed to exist, or recognized as occurring, in two stratigraphic time units will be assigned the number of the overlying (most recent) stratigraphic unit of the two.

AAPG Time-Stratigraphic Code - Columns 72-74

The numbers assigned are those in the following AAPG Code:

Cenozoic	100	Paleozoic	300	Paleozoic--Cont.	
Quaternary	110	Permian	310	Devonian	340
Recent	111	Upper	311	Upper	341
Pleistocene	112	Ochoa	312	Chautauquan	342
		Guadalupe	313	Senecan	343
Tertiary	120	Lower	317	Middle	344
Pliocene	121	Leonard	318	Erian	345
Miocene	122	Wolfcamp	319	Lower	347
Oligocene	123			Ulsterian	348
Eocene	124	Pennsylvanian	320		
Paleocene	125	Upper	321	Silurian	350
		Virgil	322	Upper	351
Mesozoic	200	Missouri	323	Cayugan	352
Cretaceous	210	Middle	324	Middle	354
Upper	211	Des Moines	325	Niagaran	355
Gulf	212	Atoka	326	Lower	
Lower	217	Lower	327	Alexandrian	358
Comanche	218	Morrow	328		
Coahuila	219			Ordovician	360
		Mississippian	330	Upper	361
Jurassic	220	Upper	331	Cincinnatian	362
Upper	221	Chester	332	Middle	364
Middle	224	Meramec	333	Champlainian	365
Lower	227	Lower	337	Lower	367
		Osage	338	Canadian	368
Triassic	230	Kinderhook	339		
Upper	231			Cambrian	370
Middle	234			Upper	371
Lower	237			Middle	374
				Lower	377
				Precambrian	400

AAPG Mnemonic Code - Columns 75-78

Alphabetic code for the formation name entered in columns 48-70 derived in accordance with the following AAPG procedure.

Words denoting geologic rank were deleted (group, formation, etc.); insignificant words were deleted (a, on, the, etc.); and qualifying words were deleted (upper, first, late, A Zone, etc.).

All non-alpha characters were replaced with blanks.

If the name contained more than four letters it was shortened until four letters remained by deleting letters according to the following procedure:

- (1) The first letter of each name was not deleted.
- (2) Letters were deleted, from right to left, in the following order (A first, E second, etc.)

A, E, I, O, U, W, H, Y, Double letter (delete one)
 T, N, S, R, L, D, C, M, F
 G, P, K, B, V, X, J, Q, Z

Words originally containing less than four letters were entered in the left and any blanks are in the right of the field.

AAPG Modification Code - Column 79

For formation names modified by number (limited to numbers from 1 to 26), or letter, an alphabetical entry is made; A for A or 1, B for B or 2, C for C or 3, etc.

Examples of use of AAPG Code and modifying code in column 71:

Example 1

Literal formation name (columns 48-70), CHEROKEE
 No entry in column 71 as "CHEROKEE" is not an AAPG Code
 AAPG Code No. (columns 72-74), 325 for Des Moines, Middle
 Pennsylvanian.

If the cards containing these data were programed for printout of geological information a check of column 71 would be included. No entry in column 71 would indicate no redundancy if both the literal formation name and AAPG Code name corresponding to 325 are printed. Therefore, the program

would print both CHEROKEE (as read from the card) and DES MOINES, MIDDLE PENNSYLVANIAN (from a name list of AAPG Codes required as an integral part of the program).

Printout of this example would be satisfactory without use of the code in column 71 as both the literal geological name and AAPG Code name are needed in the printout.

Example 2

Literal formation name (columns 48-70), DES MOINES

An entry of 3 is in column 71 as DES MOINES is an AAPG Code for which a number is assigned.

AAPG Code No. (columns 72-74), 325

In the program for this example, a check on column 71 would indicate redundancy if both the literal name and the AAPG code name for 325 are used. Only the AAPG Code name would be programed for printout: DES MOINES, MIDDLE PENNSYLVANIAN.

Without the differentiation provided by the code in column 71, printout of the literal geological name and AAPG Code name would result in:

DES MOINES, DES MOINES, MIDDLE
PENNSYLVANIAN

This redundancy could have been avoided without column 71 by making no entry of AAPG Code name in the literal formation name field, i.e., columns 48-70 would contain no entry when only an AAPG Code name was known. However, this would not have permitted satisfactory use of the cards, without programing, for simple read and print operation.

Example 3

Literal formation name (columns 48-70), MIOCENE and information available indicates LOWER.

An entry of 8 is in column 71 as MIOCENE is an AAPG Code.

AAPG Code No. (columns 72-74), 122

In the program for this example, a check of column 71 would indicate two things: (1) Redundancy if both the literal name and the AAPG Code name for 122 were printed and (2) the printout should include LOWER. Printout would be programmed as LOWER MIOCENE.

AAPG Code for samples from two or more formations

If the names of two or more formations (or the word composite) are entered as literal formation name then CMPS, the mnemonic code for composite, is entered in columns 75-78.

If the stratigraphic location of all the formations entered is such that one AAPG code No. applies to all, then that No. is entered in columns 72-74; if not, 000 (zeros) is entered.

APPENDIX

Alpha and Numeric Codes for Oil Producing Countries
and Their Political and Geographical Subdivisions

AFRICA, Northern				ASIA, Far East (cont'd)			
Algeria	E1			Taiwan (Formosa)	P3		
Oran area	E1	013		Shinchiku	P3	005	
Sahara	E1	015		Thailand	P7		
Egypt (UAR)	E3			Chiengmai	P7	001	
Sinai	E3	007					
Libya	E5			ASIA, Middle East			
Tripolitania	E5	001		<i>Abu Dhabi</i>	K1		
Cyrenaica	E5	003		Bahrain (Bahrein)	K1		
Morocco	E6			Iran	K2		
				Khuzistan	K2	003	
AFRICA, Western and Southern				Iraq	K3		
Angola	E2			Kirkuk area	K3	001	
Nigeria	E4			Israel	K4		
Malagasy Republic	E8			Kuwait	K5		
(Madagascar)				Saudi Arabia	K8		
Majunga	E8	001		Eastern portion	K8	003	
				Forsan Islands	K8	005	
ASIA, Far East							
Burma	P1			EUROPE, Eastern			
India	P4			Albania	H1		
Assam	P4	001		Czechoslovakia	H5		
Japan	P5			Moravia	H5	003	
Akita	P5	003		Slovakia	H5	005	
Niigata	P5	009		Greece	H8		
Ishikawa	P5	035		Zante (Zakynthos)	H8	049	
Hokkaido	P5	045					
Pakistan	P6						

EUROPE, Eastern (cont'd)

Rumania (Romania)	J4	
Moldavia	J4	007
Muntenia	J4	009
U.S.S.R. (Russia) . . M1 and N1		
Azerbaidzhan	M1	005
Kazakh	N1	023
Russian Soviet Federated		
Socialist Republic	M1	100
Chusovoy area	M1	111
Syzran area	M1	151
Sakhalin Island	N1	037
Tadzhik	N1	041
Ukraine	M1	045
Yugoslavia	J6	

EUROPE, Western

France	H6	
Bas Rhin	H6	025
Herault	H6	087
Germany, West	H7	
Hannover	H7	007
Italy	J1	
Emilia	J1	009
Netherlands	J2	
Norway	J3	150
Northern	J3	150
United Kingdom	H3	
Scotland	H3	140
North Sea	H3	150

NORTH AMERICA

Dominion of Canada		
Alberta	61	
Medicine Hat	61	001
Lethbridge	61	002
Macleod	61	003

NORTH AMERICA (cont'd)

Dominion of Canada (cont'd)

Alberta (cont'd)	61	
Acadia	61	004
Bow River	61	005
Calgary	61	006
Battle River	61	007
Red Deer	61	008
Camrose	61	010
Wetaskiwin	61	011
Vegreville	61	012
Athabaska	61	013
Peace River	61	015
Jasper Edson	61	017

British Columbia	62	
Division 10 (Ft.		
St. John)	62	010

Manitoba	63	
Souris	63	004
Brandon	63	007

New Brunswick	64	
Albert	64	008

Newfoundland	65	
Division 4 (St. Georges-		
Port au Port)	65	004

Northwest Territories . .	66	
McKenzie (Ft. Norman		
area)	66	003

Ontario	68	
Brant	68	002
Elgin	68	009
Haldimand	68	015
Kent	68	020
Lambton	68	021
Manitoulin	68	026
Middlesex	68	027
Norfolk	68	030
Simcoe	68	043

NORTH AMERICA (cont'd)

Dominion of Canada (cont'd)		
Quebec	70	
Gaspe	70	020
Saskatchewan	71	
Assniboia	71	001
Weyburn	71	002
Wood Mountain	71	003
Maple Creek	71	004
Qu'Appelle	71	005
Swift Current	71	008
Battleford	71	013
Mexico	A3	
Nueva Leon	A3	033
Oaxaca	A3	035
Tabasco	A3	049
Tamaulipas	A3	051
Veracruz	A3	055
United States of America		
(See "Numerical Code for States, Counties, and Cities of the United States," by International Business Machines Corp., 1952 and 1958, 104 pp.)		
Alaska	50	
North of Brooks Range	50	100
Barrow Point area	50	105
Cape Simpson	50	111
Fish Creek	50	127
Umiat area	50	187
Wolf Creek	50	195
South of Alaska Range	50	300
Cook Inlet area	50	331
Katalla area	50	365
Louisiana	17	
Offshore	17	900

OCEANIA

Australia	V1	
Victoria	V1	011
Queensland	V1	005
Western	V1	013
Indonesia	S1	
Borneo (Kalimantan)	S1	
Tarakan	S1	003
Java	S4	
Sumatra	S7	
North New Zealand	V3	
New Plymouth area	V3	001
Sarawak	S1	001

SOUTH AMERICA

Argentina	C1	
Chubut	C1	005
Neuquen	C1	029
Chile	C5	
Magallanes	C5	025
Colombia	C6	
Antioquia	C6	003
Bolivar	C6	009
Boyaca	C6	011
Magdalena	C6	027
Santander	C6	035
Santander Del Norte	C6	037
Peru	C8	
Piura	C8	035
Venezuela	D1	
Delta Amacuro	D1	001
Anzoategui	D1	005
Guarico	D1	023
Zulia	D1	047
Lake Maracaibo	D1	025

WEST INDIES

Cuba	A2	
Camaguey	A2	001
Havana	A2	005
Trinidad	C9	